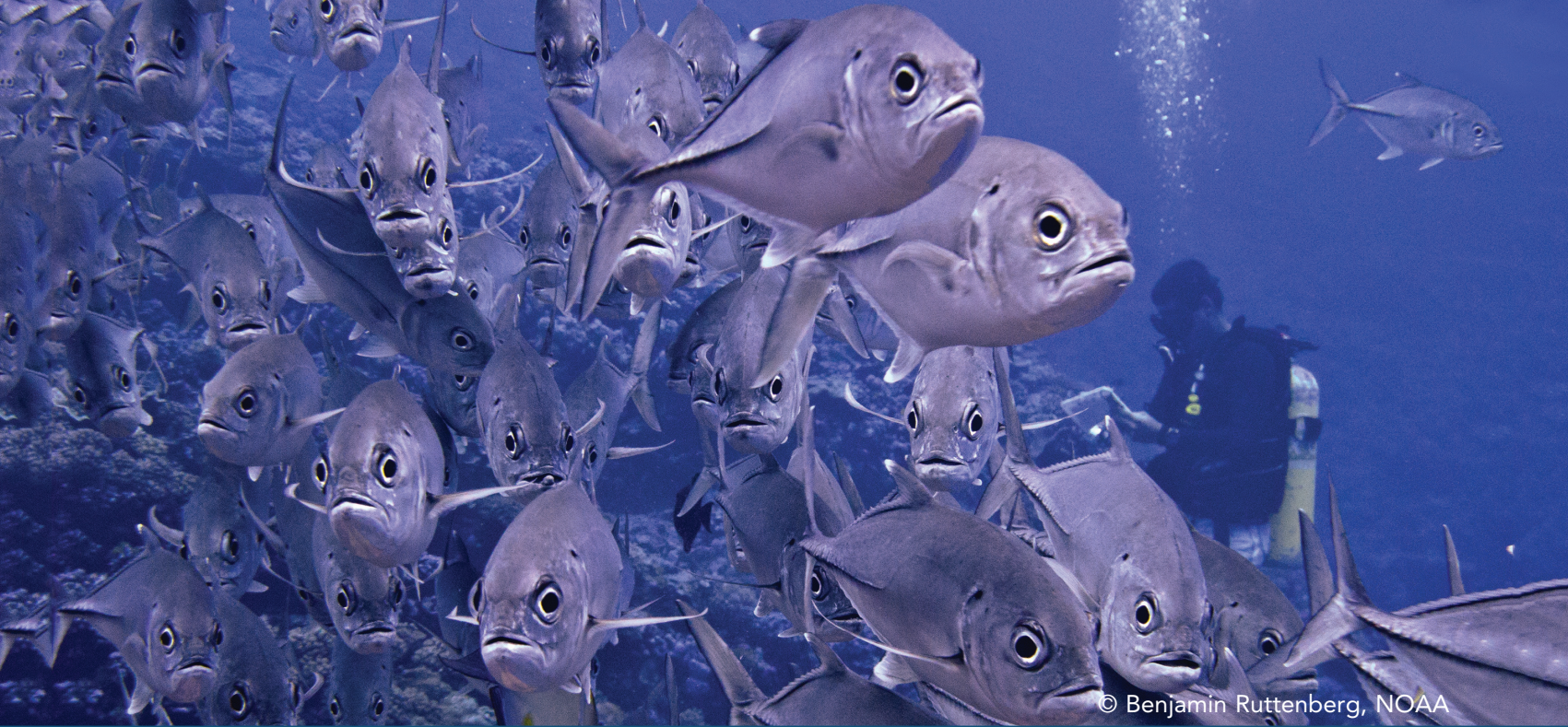

33 Quantifying reef fishes: bias in observational approaches

Paolo Usseglio

Our understanding of the dynamics of coral reef fish populations has been shaped by our ability to quantify their distribution, abundance, and body size across temporal and spatial scales. These quantifications are, however, subject to biases inherent to sampling methodologies. Here, I review scientific papers published in the last 50 years that address such biases. The most common biases include: fish behavior and observer presence, observer error, sampling method, and transect dimension. While some of these biases can be addressed with the use of distance sampling methods, observer calibration exercises, and available technology, it is important to note that identifying biases and factoring them into our resulting estimates will be fundamental to assessing the reliability of our conclusions and overall improve our understanding of reef fishes.

Ecology of Fishes on Coral Reefs, ed. C. Mora. Published by Cambridge University Press. © Cambridge University Press 2015.

https://doi.org/10.1017/CBO9781316105412.035 Published online by Cambridge University Press

BIASES QUANTIFYING REEF FISHES

The ecology of fishes on coral reefs has been developed mainly by our ability to quantify the distribution, abundance, and body size of fishes in space and time. There are, however, inherent biases in sampling methodologies that prevent an accurate and precise description of fish communities, making the study of these biases a critical first step in ecology.

Reef fishes are sampled mostly by means of underwater visual censuses (UVC). The most common sampling methods used in UVC include: belt transects (i.e. the observer counts fishes at both sides of a line deployed over the reef; e.g. [343]), stationary point counts (i.e. the stationary observer counts all fishes within a given radius of the field of view; e.g. [132]), line transects (i.e. observer records fish and their perpendicular distance to the transect; e.g. [2487]), towed censuses (i.e. observer records fish while being towed behind a boat; e.g. [700]), and rapid visual censuses (i.e. observer records species and the time at which they are seen; e.g. [319,646]).

Several sources of biases associated with these methods have been mentioned and studied in the reef fish literature, and are reviewed here. My review included a search of publications, ranging from 1954 to 2013, which specifically addressed issues related to biases in UVC. This search yielded 75 papers that identified a total of 20 sources of bias (Table 33.1). All biases are presented in Table 33.1, and the most common are briefly discussed below.

Fish behavior and observer presence

In a broad sense, this limitation is associated with the life-history traits of a given species, and how these will affect the observer's ability to detect a fish. These traits include crypsis (i.e. camouflage, countershading), motility, and flight response. Almost half of the reviewed papers recognize that UVC methods are greatly biased against small cryptic species. This limitation could be reduced with the use of lethal chemicals, such as rotenone [13], but given the destructive nature of these chemicals, this is seldom feasible and in many countries illegal [2140].

Fish behavioral response to observer presence (e.g. diver or snorkeler) can impact the number of fish that are detected during the survey, as once an observer enters the water, fish distribution becomes a function of the fish response to the observer rather than a reflection of the true density before the observer entered the water (Figure 33.1). Fishing pressure, by means of spear guns or pole spears, can induce a learning behavior that significantly modifies flight response ([803,1392]; Figure 33.1). This situation can lead to underestimated counts, as fish moving away from the observer will leave the transect boundary and not be counted [2626]. Conversely, fish can be curious towards the observer resulting in overestimated fish counts ([1392,1396,2626]; Figure 33.1).

Another factor generating attraction or flying responses in fish relates to the use of open-circuit SCUBA gear used in most UVC, which is not quiet [2056]. Although this bias can be minimized with the use of closed-circuit apparatus, such as re-breathers, they are largely inaccessible due to their high operational cost, and more demanding logistics. To minimize errors associated with fish behavior, it is often recommended to do an acclimation period, in which the diver remains still for 5–10 minutes at the start of each survey [670,848]. However, even if a given survey protocol calls for it, observers would seldom stay idle, as they are not neutral objects, and will still produce bubbles. It has also been suggested that biases associated with movement away or attraction towards the observer can be reduced by the use of instantaneous counts ([2609], but see 1396]).

Observer error

During an UVC, an observer typically estimates transect width, fish size, number of fish, and, depending on the method, distance to the fish. All these quantifiable attributes can be wrongly measured by the observer (i.e. intra-observer bias) and vary among observers (i.e. inter-observer bias). Of the papers reviewed, 25.3% identified observer error as a source of bias. Intra-observer bias has been shown to be reduced with size and count calibration exercises [174], and the use of lasers, which can project a pair of dots spaced a set distance onto a fish allowing a set of points of reference over the fish that minimizes errors in fish length estimations [1090]. In turn, inter-observer bias can be reduced with calibration exercises [619], by using experienced or well-trained observers, the same observers [2685], or by using underwater video cameras [e.g. 310]. While underwater video systems are more accurate and precise than divers in terms of abundance and size estimation [310], time commitment in processing of images, and data storage can become quite significant.

Sampling method and transect dimension

In the reviewed literature the most commonly used methods for UVC are belt transects and stationary point counts. While some suggest that these methods yield comparable estimates [2251], others report substantial differences in estimates [527,551]. Higher observer attention at the beginning of a census has been suggested to account for the higher abundances and diversity yielded by stationary point counts [1396].

Transect dimension has been shown to influence estimates of abundance and diversity. Sale and Sharp [2228] found, for instance, that density estimates drop as transect width increases. This pattern can be explained by the detection function of a given object, whereby the further an object is from an observer the lower the probability of seeing it. This detection function is a factor of variables such as fish size, fish numbers, color, crypsis, etc. (Figure 33.1). This bias can be reduced with the use of distance sampling [374], which allows the quantification of

https://doi.org/10.1017/CBO9781316105412.035 Published online by Cambridge University Press

Table 33.1 Sources of bias associated with underwater visual censuses (UVC) in a review of 75 peer-reviewed journals from 1954–2013.

Source of variation	References	Papers (#)	Papers (%)
Fish behavior (detectability, crypsis, motility, flight response)	[13,171,281,312,343,394,473,539,736,763,803,848,985,1242,1319,1390, 1392,1393,1396,1397,1534,1551,1716,1944,1962,2032,2251,2478, 2487,2625-2627,2694,2695,2697]	35	48.0
Observer error (distance, size estimates, inter-observer variability)	[222,310,343,619,736,848,1027,1051,1052,1417,1944,2196,2254,2432, 2433,2477,2478,2487,2685]	19	25.3
Sampling technique	[299,527,539,551,646,1052,1393,1396,1534,1601,2227,2376,2627,2694]	14	18.7
Observer presence (SCUBA noise, diver presence)	[222,434,640,670,1315,1392,1601,1943,1944,2056,2102,2196,2478]	13	17.3
Transect dimension (length, width)	[312,319,439,848,1396,1648,2228,2487]	8	10.7
Spatial distribution and abundance of target species	[646,848,1339,1648,1962]	5	6.7
Water clarity	[312,1339,2487,2694]	4	5.3
Sampling time (diver speed)	[2032,2251,2376,2432]	4	5.3
Habitat complexity	[375,848,985]	3	4.0
Non-instantaneous counts, instantaneous variation	[1210,1624,2609]	3	4.0
Number of replicates	[848,2227]	2	2.7
Fish speed	[1339,2487]	2	2.7
Number of species counted	[2196,2376]	2	2.7
Edge effect	[77,2473]	2	2.7
Observer speed	[2376]	1	1.3
Minimum size of fish sampled	[182]	1	1.3
Census time interval	[1339]	1	1.3
Recording media (slate vs. audio)	[299]	1	1.3
SCUBA constraints (weather, depth, daylight)	[1242]	1	1.3
Spatial coverage of surveys	[2102]	1	1.3

detectability, or the probability of detecting an object of interest. This idea is founded in the principle that the farther away from the observer an object of interest is, the lower its chances of being sighted (see [1406] for application to UVCs).

DISTANCE SAMPLING AS A WAY TO ADDRESS SOME BIASES

To date fish life-history traits, fish reaction to observers, and the ability of an observer to detect a fish are sources of bias that remain largely overlooked in the use of UVC. Our terrestrial counterparts, however, have taken into consideration unaccounted individuals in their estimates of abundance with the use of distance sampling [374].

By recording the distance of the object of interest to the observer it is possible to calculate the probability of detection of the object. Most UVC methods assume that observers do not overlook their target, but this can be difficult; thus, detectability functions adjust abundance estimates leading to increased accuracy. Such a method could also help minimize the extent of

https://doi.org/10.1017/CBO9781316105412.035 Published online by Cambridge University Press

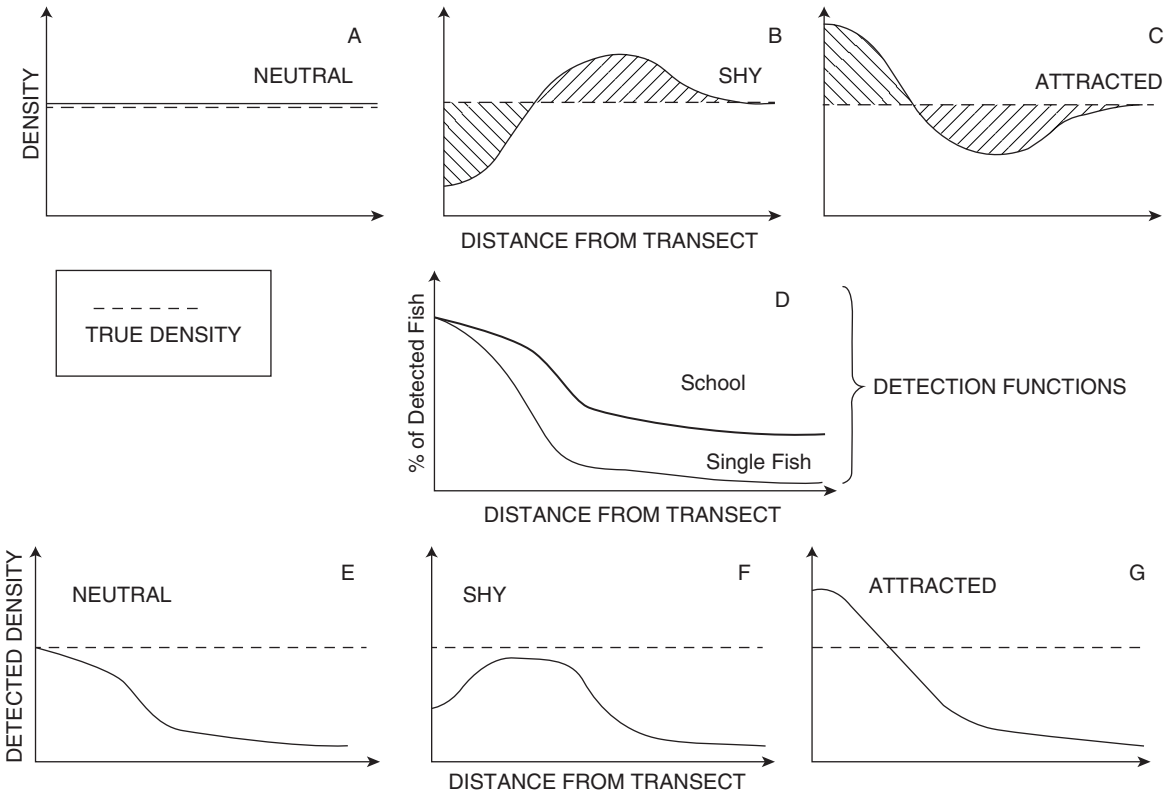

Figure 33.1 Changes in the spatial distribution of densities of fish once the observer is in the water. (A) Before the observer gets in the water, species do not change position, and thus density is not related to observer and can be assumed to be equal over the range to be sampled. (B) Reaction of shy species, moving away from the observer, density drops close to the observer and increases away from the observer. (C) Reaction of attracted species, moving closer to the observer, density increases close to the observer but decreases at some distance from the observer. (D) Detection curve or the probability of detecting a fish, or school, as a function of distance from transect (see distance sampling theory). (E–G) Fish available to count by an observer resulting from combining detection functions (D) and actual density of fish (A–C) for neutral (E), shy (F), and attracted (G) species. Distance sampling theory estimates density prior to the presence of an observer by combining the detection curves and the final observations – counts. Image courtesy of M. Kulbicki.

knowledge biases. For instance, biases associated with flight response could be minimized as such behaviors will result in peaks in the detection functions that are shifted away from the observer (Figure 33.1F; Bozec [312]). Sampling distance could also minimize the extent of biases related to fish detectability mediated by abundance, size, and topographic complexity (Figure 33.1; [312,1392]). While distance sampling can address several of the sampling biases identified in this review, it is important to stress Kulbicki's [1392] warning that distance sampling is no panacea.

CONCLUDING REMARKS

Some of the biases associated with UVC can be reduced through training (e.g. improving underwater estimations of fish size and length) and available technology (e.g. laser calipers and cameras), but some other biases cannot (e.g. fish attraction to the observer). This highlights the importance of identifying sampling biases and factoring them into resulting estimates. This practice will provide margins of error in the conclusions or give an insight in terms of the reliability of these conclusions. Increased focus on the sampling biases of UVC will advance our ability to quantify the distribution and abundance of fishes and by default improve our understanding of this very diverse taxa.

https://doi.org/10.1017/CBO9781316105412.035 Published online by Cambridge University Press